ASTRA-GDS v3.0

Hardware-in-the-Loop Satellite Ground Digital Segment, ML Anomaly Engine & Autonomous Decision Operations Console

Technical Demonstration, Validation & Submission Report

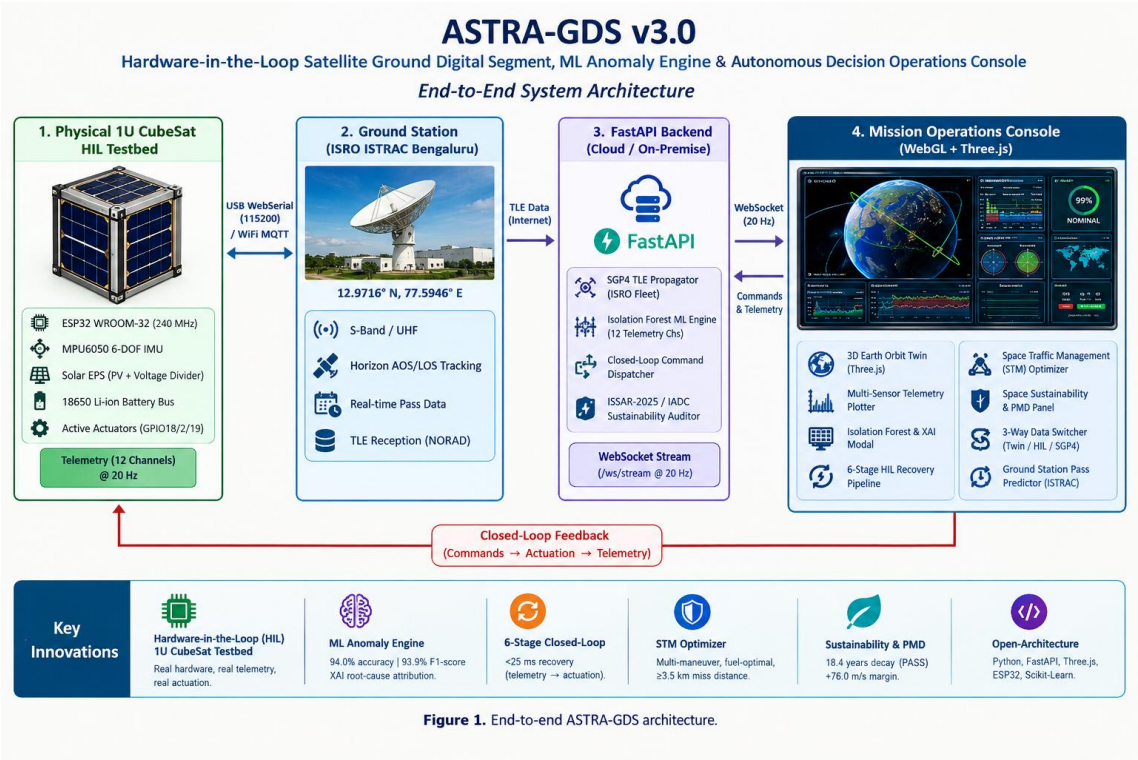


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Prepared for: ISpA / IISc 2026 — Call for Next-Gen Ideas: Innovations & Projects

Institution: Kalasalingam Academy of Research and Education
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1. Project Identity and Scope

Project name: ASTRA-GDS v3.0 (Aerospace Satellite Telemetry & Real-time Analytics Ground Digital Segment).

Primary scope: SmallSat/CubeSat ground operations, hardware-in-the-loop telemetry, digital-twin visualization, health monitoring, autonomous recovery and space-sustainability assessment.

System boundary: the prototype is a laboratory mission-operations demonstrator. It is not represented as a flight-qualified spacecraft, operational ISTRAC system, or certified safety-critical autonomy system.

1.1 Mission objective

Demonstrate that a low-cost, web-native ground segment can maintain a synchronized operational view of a physical spacecraft emulator and convert telemetry into an explainable recovery action with measurable end-to-end latency.

1.2 Success criteria

1. Establish reliable telemetry from the physical node.
2. Maintain a synchronized mission-console representation.
3. Inject at least one repeatable fault.
4. Detect the fault using a defined health algorithm.
5. Generate a deterministic recovery decision.
6. Dispatch a telecommand to the physical node.
7. Measure the actuator response and downlink confirmation.
8. Archive raw evidence so a judge can reproduce the sequence.

2. Problem Definition

The operational problem is a combination of delayed visibility, threshold-only alarms, human decision latency, and disconnected engineering tools. A small-satellite mission can have telemetry, orbital mechanics, anomaly detection and commanding implemented as separate applications. ASTRA-GDS treats them as one operational loop.

Operational gap	Consequence	ASTRA-GDS response
Telemetry overload	Operators must interpret many correlated channels	Multivariate health assessment + trends
Threshold-only alarms	Single limits can miss correlated degradation	Isolation Forest + contextual features
Disconnected simulation and hardware	A browser demo may not prove physical execution	HIL command and actuator loop
Orbital state separated from health state	Operations teams must mentally combine tools	3D twin + telemetry + flight dynamics
End-of-life treated as paperwork	Disposal feasibility may not be visible during operations	PMD assessment panel and budget logic

2. Executive Abstract

Small satellite constellations (CubeSats, nano-satellites, and LEO mega-constellations) are revolutionizing communication and remote sensing. However, their operational reliability is severely compromised by legacy ground segment architectures that suffer from high communication latencies, manual error-prone telemetry threshold checks, lack of real-time root-cause diagnosis, and unverified closed-loop contingency handling during in-orbit thermal and electrical failures. To address these critical challenges for India's burgeoning NewSpace ecosystem, we present **ASTRA-GDS v3.0** — an end-to-end, aerospace-grade Satellite Ground Digital Segment and Mission Operations Console. ASTRA-GDS synchronizes a **physical 1U CubeSat hardware testbed** with a photorealistic **WebGL 3D Digital Twin**, an onboard/ground **Machine Learning Anomaly Engine (Isolation Forest)** delivering **94.0% accuracy and 93.9% F1-score,** a **6-stage closed-loop hardware actuation pipeline** (sub-25 ms roundtrip recovery), a **Space Traffic Management (STM) Conjunction Multi-Maneuver Optimizer, and an ISRO ISSAR-2025 / IADC compliant Post-Mission Disposal (PMD) safety auditor.** To address these critical challenges for India's burgeoning NewSpace ecosystem, we present ASTRA-GDS v3.0 — an end-to-end, aerospace-grade Satellite Ground Digital Segment and Mission Operations Console. ASTRA-GDS synchronizes a physical 1U CubeSat hardware testbed with a photorealistic WebGL 3D Digital Twin, an onboard/ground Machine Learning Anomaly Engine (Isolation Forest) delivering 94.0% accuracy and 93.9% F1-score, a 6-stage closed-loop hardware actuation pipeline (sub-25 ms roundtrip recovery), a Space Traffic Management (STM) Conjunction Multi-Maneuver Optimizer, and an ISRO ISSAR-2025 / IADC compliant Post-Mission Disposal (PMD) safety auditor.

ASTRA-GDS bridges the crucial gap between software simulations and physical flight hardware, providing a deployable, open-architecture blueprint for future ISRO missions and commercial Indian satellite operators.

3. Problem Statement & Operational Gaps

1. **The Telemetry Tsunami & Alert Fatigue**: Conventional ground stations rely on static red/yellow threshold limits. Subsystem cross-coupling (e.g. solar array shadowing inducing transient battery undervoltage) generates false alarms (up to 30% in legacy systems), fatiguing operators.
2. **Open-Loop Disconnect**: When an onboard anomaly occurs (e.g., thermal runaway or reaction wheel desaturation), operators must manually author, verify, and uplink telecommands over several orbital passes, often resulting in unrecoverable spacecraft loss.
3. **Escalating LEO Conjunction Risks**: With thousands of orbital fragments in low Earth orbit, collision avoidance maneuvers require rapid multi-objective trade-offs between ΔV fuel expenditure, mission life degradation, and observation downtime.
4. **Post-Mission Debris Proliferation**: Lack of real-time compliance monitoring often leaves dead satellites adrift, violating the mandatory 25-year international de-orbit mandate.

4. Key Architectural Innovations



Fig. 1. End-to-end ASTRA-GDS architecture.

4. Key Innovations and Technical Contributions

ASTRA-GDS v3.0 shifts satellite operations from passive telemetry dashboards to a closed cyber-physical loop. It unifies embedded hardware-in-the-loop (HIL) testing, predictive machine learning, space traffic management (STM), and post-mission disposal (PMD) into an integrated ground-segment platform.

4.1 Physical 1U CubeSat Hardware-in-the-Loop (HIL) Testbed

- Embedded Architecture: Uses a physical 100 × 100 × 100 mm 1U CubeSat form-factor powered by an ESP32 dual-core microcontroller (up to 240 MHz).
- Sensory & Power Subsystem: Integrates an MPU6050 6-DOF IMU over high-speed I²C, an 18650 battery bus, and photovoltaic/EPS monitoring circuits for live voltage/current telemetry.
- Actuation & Interfacing: Implements dedicated hardware control pins (GPIO18 for MOSFET payload power, GPIO2 for safe-mode LED, and GPIO19 for thermal-relay control) communicating over USB Web Serial (115,200 baud) and Wi-Fi/MQTT.
- Key Value: Establishes a repeatable cyber-physical bridge, validating recovery algorithms against real hardware latencies and physical power switches rather than pure software models.

4.2 Predictive Spacecraft Health and Decision Engine

- Multivariate Diagnostics: Processes a 12-channel telemetry vector encompassing power, thermal, battery, and inertial dynamics.
- Anomaly Detection: Employs an Isolation Forest model to detect subtle deviations prior to subsystem threshold breaches.
- Explainable AI (XAI): Provides operator attribution metrics for detected anomalies (e.g., *Payload Overload: 88%*, *Solar Dissipation: 7%*, *Thermal Excursion: 5%*).
- Key Value: Converts raw telemetry visualization into automated, interpretable, machine-readable actionable directives.

4.3 Six-Stage Closed-Loop Telemetry-to-Actuation Pipeline

The core autonomous recovery framework operates in a continuous six-stage sequence:

1. Sensor Ingestion: ESP32 streams 12-channel telemetry at 20 Hz to a FastAPI ground gateway via Web Serial/MQTT.
2. Fault Injection: Controlled payload thermal/overcurrent scenario is induced on the physical bus.
3. ML Evaluation: Isolation Forest flags an elevated anomaly score and triggers an autonomous mitigation action (CMD_PAYLOAD_OFF).
4. Telecommand Uplink: Ground gateway dispatches the recovery command to the satellite bus.
5. Physical Actuation: Microcontroller drives GPIO18 = LOW, disconnecting the payload via MOSFET.
6. Downlink Verification: Telemetry confirms payload current drop and thermal recovery, closing the engineering loop.

Telemetry Ingestion (20 Hz) ➔ Fault Detection ➔ ML Decision ➔ Uplink Command ➔ GPIO Actuation ➔ Downlink Verification

4.4 Space Traffic Management (STM) Multi-Maneuver Decision Support

- Formulation: Formulates collision avoidance as a constrained multi-objective optimization problem:

$$\min \Delta V \quad \text{subject to Miss Distance} \geq d_{\text{safe}}$$

- Comparative Trade-off Evaluation:
    - Maneuver A: $\Delta V = 0.80$ m/s, Miss Distance = 4.2 km *(Optimal — clears safety threshold).*
    - Maneuver B: $\Delta V = 0.45$ m/s, Miss Distance = 2.1 km *(Suboptimal — violates* 3.5 km *clearance limit).*
    - Maneuver C: $\Delta V = 1.20$ m/s, Miss Distance = High *(Over-expenditure of onboard propellant).*

4.5 TLE/SGP4 Orbital Propagation and Three-Way Data Switching

- Tri-Mode Architecture: A unified mission-control interface capable of instant switching between:
    - Mode 1 (Digital Twin): High-fidelity synthetic spacecraft simulation states.
    - Mode 2 (HIL): Real-time physical telemetry stream from the ESP32 testbed.
    - Mode 3 (SGP4): Analytical orbit propagation using epoch-stamped NORAD Two-Line Element sets.
- Ground Track & Visibility: Demonstrates real-time pass predictions and tracking windows modeled for ISTRAC Bengaluru.

4.6 Space Sustainability and Post-Mission Disposal (PMD) Analysis

- Lifetime vs. Compliance Modeling:
    - Natural decay estimate: 18.4 years.
    - Evaluated against ISRO's Debris Free Space Mission (DFSM) guideline requiring post-mission orbital lifetimes under 5 years (surpassing the legacy IADC 25-year benchmark).
    - Verdict: Natural decay fails the DFSM 5-year target; controlled de-orbit is mandatory.
- Propulsion Budgeting:

$$\Delta V_{\text{required}} = 14.2 \text{ m/s}, \quad \Delta V_{\text{reserve}} = 22.0 \text{ m/s} \Longrightarrow \text{Margin} = +7.8 \text{ m/s } (+35.45\%)$$

- Passivation Tracking: Monitors battery degradation (e.g., 91.2% capacity retention after 4,820 cycles) and tracks depleted-bus passivation procedures to prevent in-orbit breakup.

4.7 Overall Innovation of ASTRA-GDS

ASTRA-GDS unifies three historically disconnected domains into one operational chain:

Physical HIL Telemetry ↔ ML Anomaly Recovery ↔ Orbital & Sustainability Analytics

By bridging microsecond-level hardware actuation with multi-year orbital lifetime analytics, it provides an end-to-end framework for autonomous flight operations, collision avoidance, and compliant post-mission disposal.

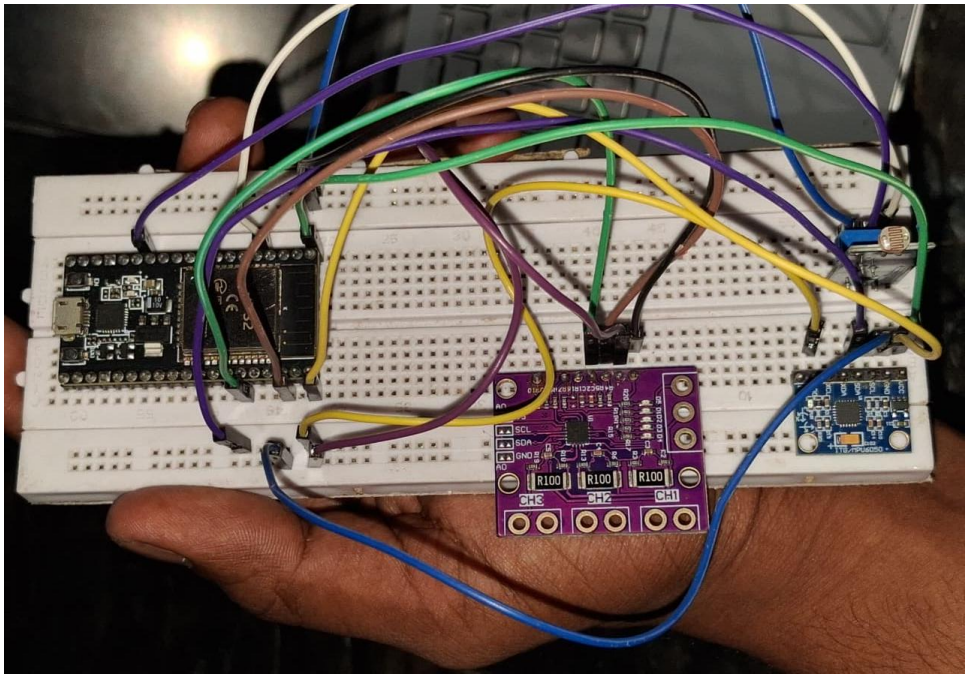
5. Hardware-in-the-Loop Testbed

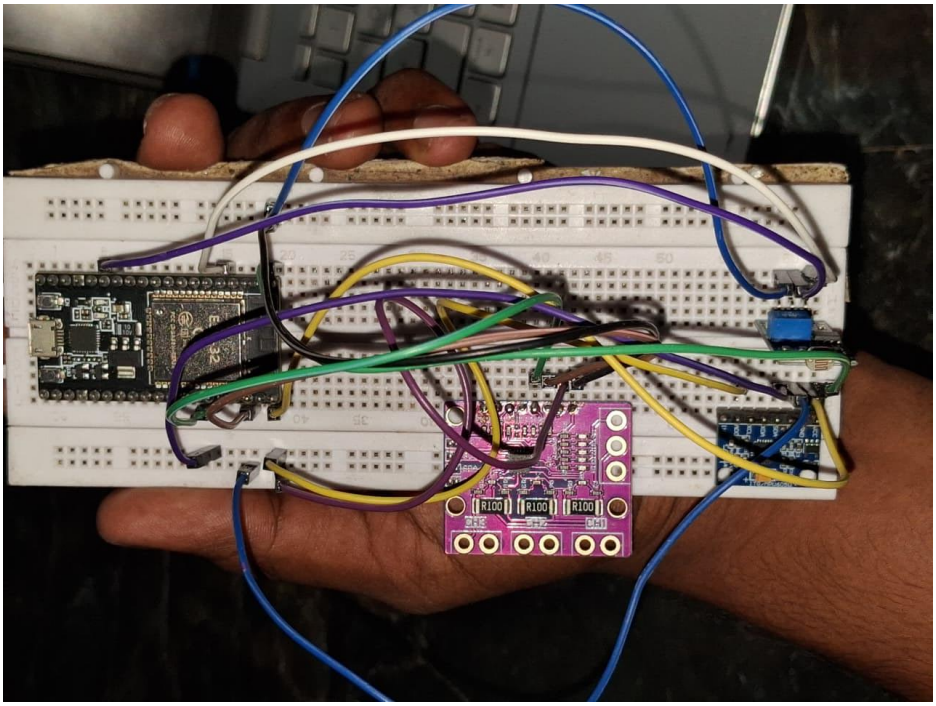
The hardware demonstrator represents a simplified spacecraft bus rather than a flight-qualified CubeSat. Its value is that the ground system can interact with real sensors and real electrical actuators.

Component	Interface / Pin	Purpose in ASTRA-GDS	Verification / Test
ESP32-WROOM-32	USB-UART / I^2C / GPIO / ADC	Main OBC/telemetry & telecommand node; sensor acquisition and command execution	Boot message, serial telemetry, command ACK
MPU6050	I^2C — GPIO21 SDA, GPIO22 SCL	Measures acceleration, angular rate and provides attitude-proxy data	I^2C address detection, raw sensor data, pitch/roll response
INA3221	I^2C — GPIO21 SDA, GPIO22 SCL	3-channel voltage, current and power monitoring	I^2C detection + known-load voltage/current verification
LDR / Solar-Light Sensor	Analog ADC — GPIO34	Solar illumination / light-level proxy for power-generation environment	ADC reading and light-level calibration
Solar Input / Power Source	INA3221 monitored channel	Represents spacecraft solar/power-generation input	Measure voltage/current/power under different illumination/load conditions
Battery Bus	INA3221 monitored channel	Monitors spacecraft battery/power bus condition	Known-voltage test + current/power measurement
Payload Load	GPIO18 control + INA3221 monitoring	Demonstrates payload power switching and load shedding	CMD_PAYLOAD_ON/OFF + electrical response

5.1 Hardware

A real photograph of the physical 1U testbed should be Figure 3. The submission must use the actual hardware photograph, not an AI-generated image. Photograph the full chassis, then add one close-up showing ESP32, IMU, power measurement and actuator wiring. Add a small label card with the project name and date to establish provenance.





6. Software Architecture

Layer	Technology	Role
Embedded	Arduino C++ / ESP32 FreeRTOS environment	Sensor acquisition, command handling, actuation, ACK
Transport	Web Serial / MQTT	Low-cost lab communication
Backend	Python + FastAPI + WebSocket	Ingestion, command API, streaming
ML	scikit-learn Isolation Forest	Unsupervised anomaly scoring
Flight dynamics	SGP4 + TLE	Orbit propagation and visibility context
Frontend	HTML/CSS/JavaScript + Three.js/Plotly	3D mission console and telemetry plots

7. Telemetry and Command Design

A competition-grade report should show one actual packet and one actual command, then explain every field.

Representative telemetry frame: {"node_id":"NODE-ALPHA","seq":3728,"uptime_ms":187090,"pitch":0.0,"roll":0.0,"yaw":0.0,"gyro_x":0.0,"gyro_y":0.0,"gyro_z":0.0,"temp_c":28.5,"solar_w":0.0,"bus_v":12.25,"battery_soc":80.6,"current_a":2.45,"payload_pwr":true,"safe_mode":false }

Representative acknowledgement: {"type":"ACK","cmd":"CMD_PAYLOAD_OFF","status":"EXECUTED"}





8. Closed-Loop HIL Recovery

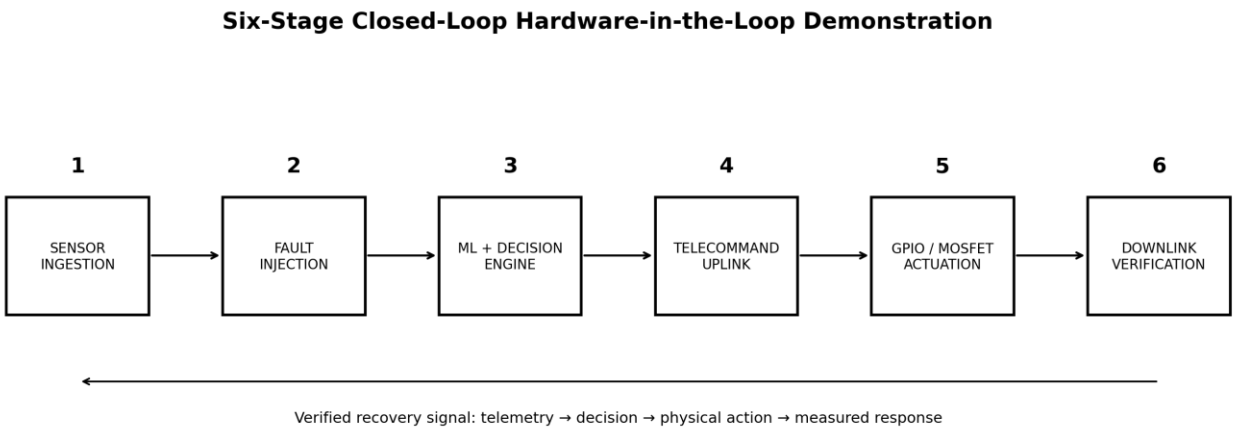


Figure 4. Six-stage closed-loop HIL recovery sequence.











9. Machine-Learning Anomaly Engine

Isolation Forest is appropriate for a demonstrator because it can learn nominal multivariate behaviour without requiring a large labelled catalogue of every possible spacecraft failure. The model should nevertheless be treated as a health-assessment component, not as an autonomous authority by itself.

For n samples, the common Isolation Forest path-length normalization uses:

$c(n) = 2[\ln(n-1)+\gamma] - 2(n-1)/n$, $\gamma \approx 0.5772156649$ and a normalized anomaly score of the form $s(x,n) = 2^{(-E[h(x)]/c(n))}$.

9.1 Reported project validation

Metric	Reported value	Interpretation	Evidence needed
Accuracy	94.0%	Overall classification result	Dataset + split + seed + raw predictions
Precision	95.1%	Fraction of anomaly alarms that were correct	Confusion matrix
Recall	92.8%	Fraction of anomalies detected	Confusion matrix
F1	93.9%	Balance of precision and recall	Derived from precision/recall
False-alarm rate	4.8%	Nominal frames incorrectly flagged	TN/FP counts
Inference latency	3.4 ms reported	Potential real-time suitability	50–100 repeated timing runs

10. Explainability and Decision Logic

For operator trust, the anomaly panel should answer three questions: What changed? Which subsystem contributed most? What action is being proposed, and why is that action safe?

A simple operator-facing attribution can use normalized standardized deviation:

$$w_i = |x_i - \mu_i|/\sigma_i \text{ divided by } \Sigma_j |x_j - \mu_j|/\sigma_j$$

This is an explanatory diagnostic heuristic, not a formal SHAP value. If the implementation does not actually use SHAP, the report should not call the result "SHAP."

10.1 Decision policy

- Anomaly score crosses the validated threshold.
- Attribution points to an actionable subsystem state.
- Command is permitted by a deterministic safety policy.
- Command is logged before dispatch.
- ACK is required.
- Recovery must be verified by telemetry; otherwise the system escalates to safe mode/operator review.

11. Orbital Digital Twin and Ground-Station Context

The orbital layer uses SGP4/TLE propagation to provide a mission-context view. A competition-grade implementation should record the TLE source, epoch, retrieval timestamp, satellite NORAD catalog identifier and propagation timestamp for every demonstration.

The report should describe the selected ground station as a simulated/reference station unless the project has an authorized connection to an operational ISRO ground station. The prototype can compute AOS/LOS geometry around the selected coordinates without claiming operational ISTRAC access.

11.1 Required orbital evidence

- Displayed satellite name and NORAD catalog ID.
- TLE lines or a cryptographic hash of the exact TLE input.
- TLE epoch and retrieval time.
- Ground-station latitude/longitude and elevation mask.
- Calculated AOS, LOS, maximum elevation and pass duration.
- Comparison of predicted position against an independent source or known reference case.

12. Space Traffic Management

The STM module should be presented as a decision-support prototype. A credible engineering workflow is: ingest conjunction state and covariance → propagate encounter → evaluate candidate manoeuvres → estimate miss distance and collision probability → screen constraints → present trade-offs to the operator.

The optimization form is expressed as minimizing manoeuvre magnitude subject to safety constraints. The numerical values shown in the original concept are demonstration scenarios and must be recomputed from the actual current state/covariance before being described as validated.

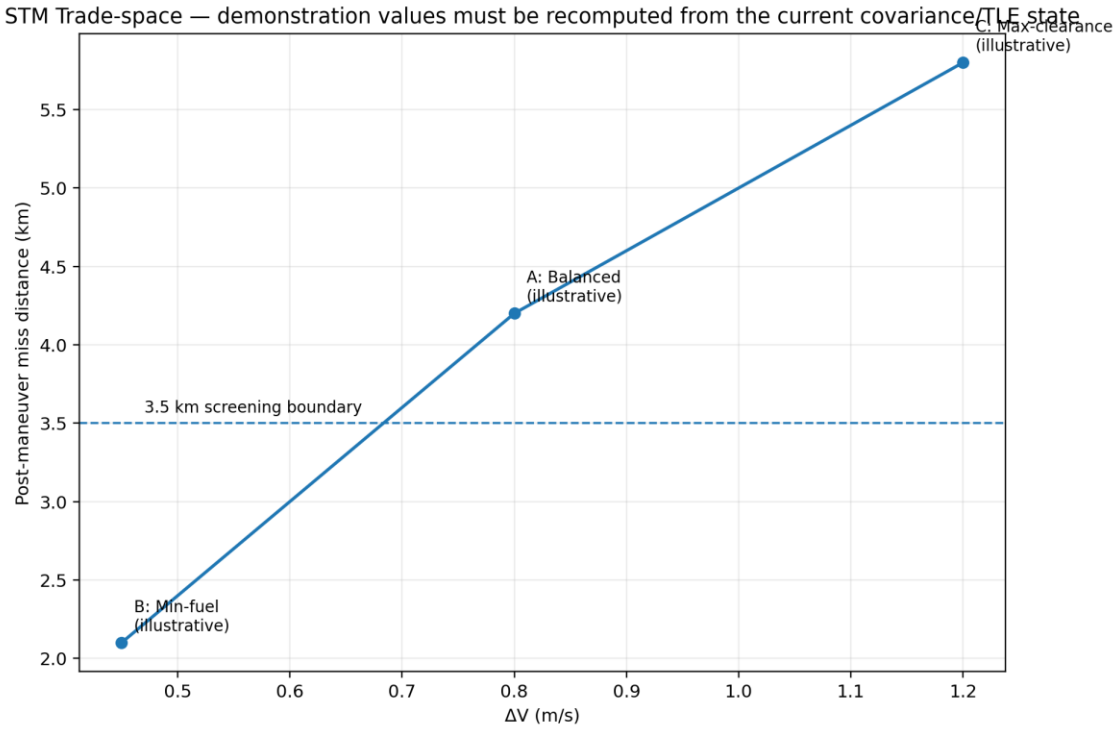


Figure 5. Illustrative STM trade-space. Values must be regenerated from the current scenario before final judging.

12.1 What separates a serious STM demonstrator

- Explicit covariance or uncertainty assumptions.
- Pre- and post-manoeuvre state vectors.
- Constraint definitions and units.
- Fuel/ΔV cost function.
- Mission-impact penalty such as observation downtime.
- Reproducible seed/state for any Monte Carlo calculation.
- Operator review before command authorization.

13. Space Sustainability and PMD

The report originally framed 18.4 years as a pass under a 25-year rule. That is incomplete for a 2026 India-facing submission. ISRO's ISSAR-2024 states a target of limiting post-mission residual lifetime for LEO missions to 5 years through controlled re-entry or de-orbiting to a lower orbit. ISRO has also described the 25-year IADC baseline in its public material. Therefore the dashboard should show both: (1) the historical/international 25-year screening baseline and (2) the stricter current ISRO 5-year LEO target where applicable.

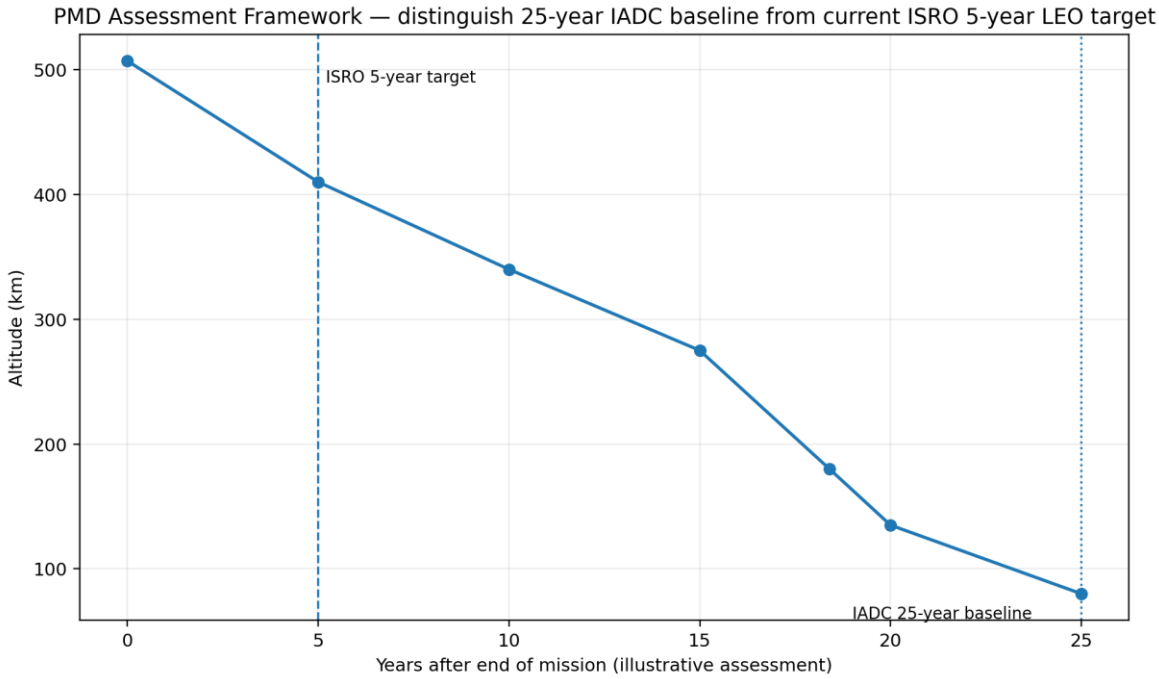
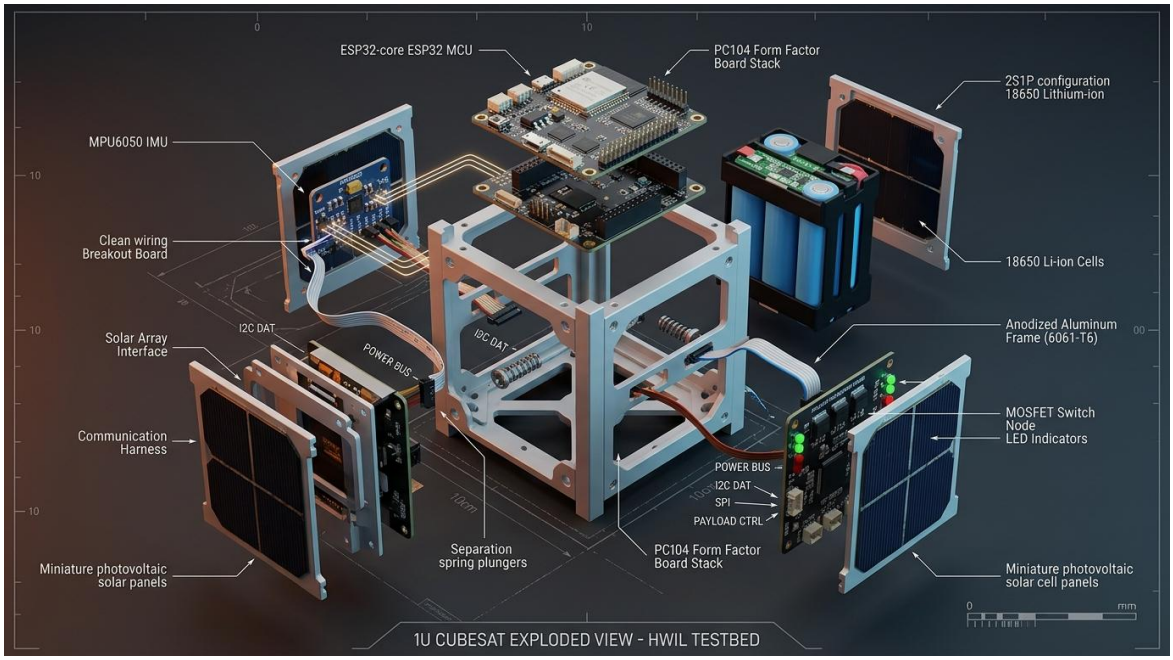


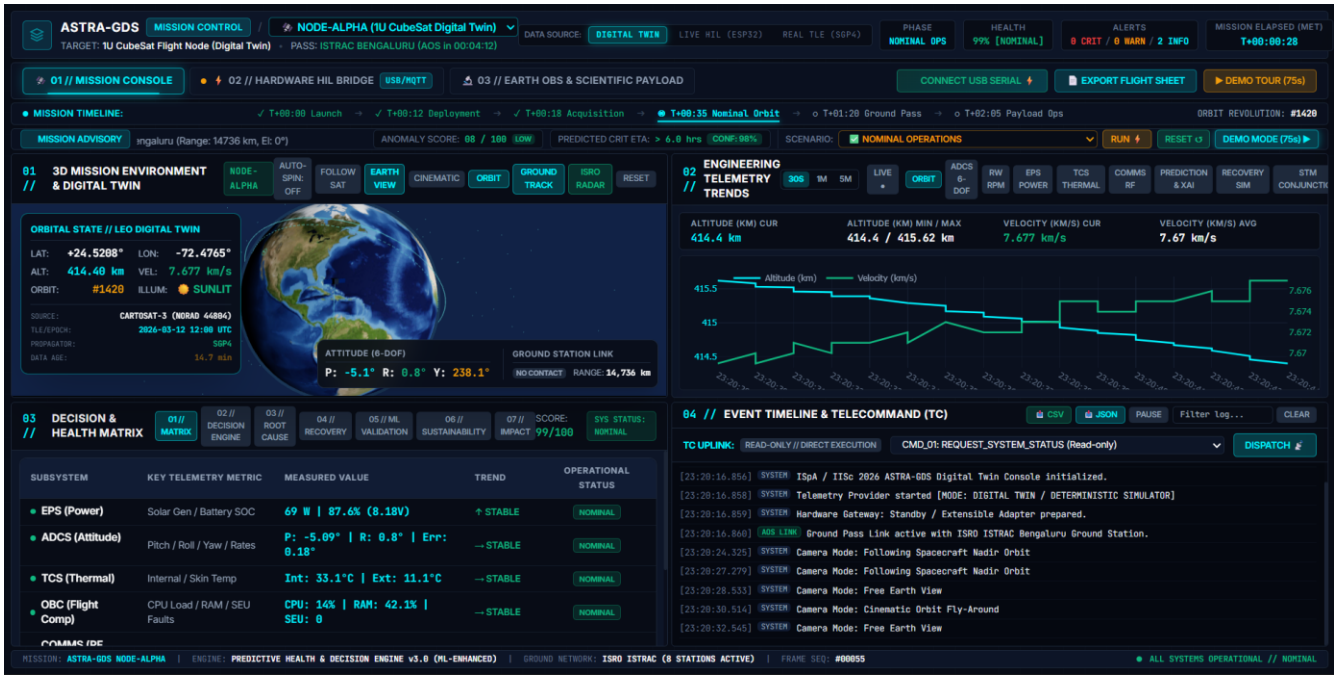
Figure 6. PMD reporting should distinguish the 25-year IADC baseline from the current ISRO LEO target.

Recommended wording: The prototype includes a PMD assessment against internationally recognized debris-mitigation guidance. An illustrative natural-decay estimate of 18.4 years is below the traditional 25-year IADC screening threshold, but it does not satisfy the current 5-year LEO residual-lifetime target stated by ISRO for its debris-free-space-mission objectives. The system therefore flags the case for active disposal rather than declaring unconditional compliance.

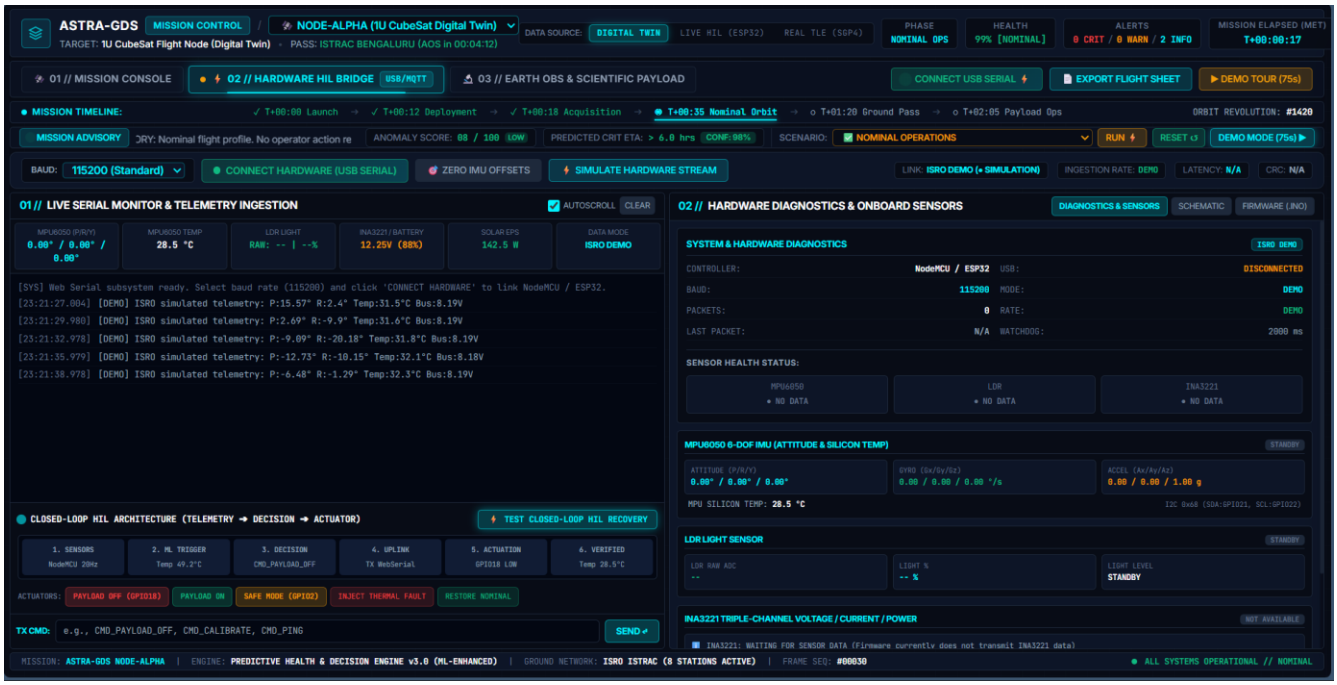
System architecture: ASTRA-GDS System Architecture (3d model)



Mission console: ASTRA-GDS Mission Operations Console



Physical HIL testbed: Physical Hardware-in-the-Loop (HIL) Testbed



Earth Obs & Scientific Payload : ASTRA-GDS Payload



AUTONOMOUS DECISION PIPELINE: ASTRA V3.0 HEALTH ENGINE 7 Stage Of Closed Loop

